

Table 3: Results of test examples 1-3

| N | Pdim  | DFPRP   |      |       |        | MPRP1    |      |       |        | RESHYBRIDSCG |      |       |        | RDFPM1   |      |       |        | RDFPM2   |      |       |        | DFDFP    |      |       |        |          |
|---|-------|---------|------|-------|--------|----------|------|-------|--------|--------------|------|-------|--------|----------|------|-------|--------|----------|------|-------|--------|----------|------|-------|--------|----------|
|   |       | IN      | FVAL | PTIME | NORM   | IN       | FVAL | PTIME | NORM   | IN           | FVAL | PTIME | NORM   | IN       | FVAL | PTIME | NORM   | IN       | FVAL | PTIME | NORM   | IN       | FVAL | PTIME | NORM   |          |
| 1 | 5000  | $x_1^4$ | 1    | 3     | 0.0077 | 0        | 67   | 135   | 0.0847 | 7.40E-11     | 21   | 43    | 0.0329 | 7.75E-11 | 11   | 17    | 0.0210 | 0        | 4    | 10    | 0.0104 | 7.95E-21 | 1    | 3     | 0.0389 | 0        |
|   | 5000  | $x_2^4$ | 1    | 3     | 0.0038 | 0        | 72   | 146   | 0.1289 | 9.24E-11     | 21   | 45    | 0.0285 | 4.97E-11 | 25   | 33    | 0.0339 | 6.49E-11 | 10   | 18    | 0.0170 | 1.30E-20 | 1    | 3     | 0.0112 | 0        |
|   | 5000  | $x_3^4$ | 1    | 3     | 0.0039 | 0        | 85   | 184   | 0.0967 | 8.88E-11     | 17   | 37    | 0.0233 | 5.42E-11 | 13   | 18    | 0.0190 | 6.88E-24 | 2    | 4     | 0.0052 | 0        | 4    | 8     | 0.0106 | 0        |
|   | 5000  | $x_4^4$ | 1    | 3     | 0.0037 | 0        | 74   | 149   | 0.0844 | 8.79E-11     | 21   | 43    | 0.0285 | 7.85E-11 | 17   | 23    | 0.0249 | 0        | 12   | 19    | 0.0187 | 2.94E-23 | 1    | 3     | 0.0033 | 0        |
|   | 5000  | $x_5^4$ | 1    | 3     | 0.0037 | 0        | 56   | 113   | 0.0653 | 8.48E-11     | 21   | 43    | 0.0272 | 7.70E-11 | 14   | 19    | 0.0209 | 1.11E-20 | 8    | 16    | 0.0160 | 2.57E-22 | 1    | 3     | 0.0036 | 0        |
|   | 5000  | $x_6^4$ | 1    | 3     | 0.0037 | 0        | 74   | 149   | 0.0870 | 8.80E-11     | 21   | 43    | 0.0305 | 7.85E-11 | 23   | 28    | 0.0298 | 0        | 12   | 19    | 0.0191 | 2.93E-23 | 1    | 3     | 0.0042 | 0        |
|   | 10000 | $x_1^4$ | 1    | 3     | 0.0052 | 0        | 67   | 135   | 0.1140 | 7.40E-11     | 21   | 45    | 0.0430 | 4.38E-11 | 6    | 12    | 0.0175 | 0        | 5    | 12    | 0.0168 | 1.68E-21 | 1    | 3     | 0.0045 | 0        |
|   | 10000 | $x_2^4$ | 1    | 3     | 0.0045 | 0        | 73   | 148   | 0.1321 | 9.32E-11     | 21   | 45    | 0.0443 | 7.03E-11 | 29   | 36    | 0.0578 | 8.72E-11 | 12   | 21    | 0.0305 | 4.29E-22 | 1    | 3     | 0.0046 | 0        |
|   | 10000 | $x_3^4$ | 1    | 3     | 0.0049 | 0        | 86   | 186   | 0.1507 | 8.96E-11     | 17   | 37    | 0.0357 | 5.42E-11 | 13   | 18    | 0.0367 | 6.88E-24 | 2    | 4     | 0.0061 | 0        | 4    | 8     | 0.0119 | 0        |
|   | 10000 | $x_4^4$ | 1    | 3     | 0.0043 | 0        | 75   | 151   | 0.1277 | 8.57E-11     | 21   | 45    | 0.0425 | 4.44E-11 | 33   | 37    | 0.0625 | 9.73E-11 | 13   | 21    | 0.0317 | 0        | 1    | 3     | 0.0044 | 0        |
|   | 10000 | $x_5^4$ | 1    | 3     | 0.0045 | 0        | 57   | 115   | 0.0996 | 8.55E-11     | 21   | 45    | 0.0469 | 4.36E-11 | 16   | 24    | 0.0385 | 2.63E-11 | 7    | 14    | 0.0197 | 0        | 1    | 3     | 0.0047 | 0        |
|   | 10000 | $x_6^4$ | 1    | 3     | 0.0049 | 0        | 75   | 151   | 0.1270 | 8.88E-11     | 21   | 45    | 0.0440 | 4.44E-11 | 20   | 27    | 0.0426 | 0        | 12   | 19    | 0.0289 | 4.11E-23 | 1    | 3     | 0.0044 | 0        |
|   | 50000 | $x_1^4$ | 1    | 3     | 0.0133 | 0        | 67   | 135   | 0.4491 | 7.40E-11     | 21   | 45    | 0.1609 | 9.80E-11 | 7    | 12    | 0.0747 | 3.70E-21 | 3    | 8     | 0.0386 | 3.57E-18 | 1    | 3     | 0.0127 | 0        |
|   | 50000 | $x_2^4$ | 1    | 3     | 0.0147 | 0        | 76   | 154   | 0.5108 | 7.57E-11     | 22   | 47    | 0.1646 | 4.40E-11 | 27   | 34    | 0.2024 | 9.25E-11 | 11   | 18    | 0.1001 | 8.28E-18 | 1    | 3     | 0.0143 | 0        |
|   | 50000 | $x_3^4$ | 1    | 3     | 0.0131 | 0        | 89   | 192   | 0.6107 | 7.28E-11     | 17   | 37    | 0.1281 | 5.42E-11 | 13   | 18    | 0.1062 | 6.88E-24 | 2    | 4     | 0.0252 | 0        | 4    | 8     | 0.0446 | 0        |
|   | 50000 | $x_4^4$ | 1    | 3     | 0.0125 | 0        | 78   | 157   | 0.5305 | 7.21E-11     | 21   | 45    | 0.1549 | 9.93E-11 | 20   | 27    | 0.1605 | 1.10E-23 | 11   | 18    | 0.1020 | 1.05E-20 | 1    | 3     | 0.0130 | 0        |
|   | 50000 | $x_5^4$ | 1    | 3     | 0.0136 | 0        | 59   | 119   | 0.3996 | 9.74E-11     | 9    | 15    | 0.1560 | 9.74E-11 | 9    | 15    | 0.0851 | 1.42E-18 | 7    | 13    | 0.0711 | 2.84E-18 | 1    | 3     | 0.0137 | 0        |
|   | 50000 | $x_6^4$ | 1    | 3     | 0.0128 | 0        | 78   | 157   | 0.5470 | 7.21E-11     | 21   | 45    | 0.1520 | 9.93E-11 | 18   | 24    | 0.1439 | 1.06E-22 | 11   | 18    | 0.0997 | 2.07E-21 | 1    | 3     | 0.0133 | 0        |
| 2 | 5000  | $x_1^4$ | 1    | 3     | 0.0046 | 0        | 63   | 131   | 0.0650 | 9.27E-11     | 19   | 41    | 0.0223 | 9.86E-11 | 13   | 21    | 0.0277 | 0        | 5    | 13    | 0.0091 | 0        | 5    | 9     | 0.0086 | 5.95E-11 |
|   | 5000  | $x_2^4$ | 1    | 4     | 0.0039 | 0        | 82   | 171   | 0.0842 | 9.26E-11     | 2    | 6     | 0.0047 | 0        | 27   | 35    | 0.0357 | 0        | 10   | 16    | 0.0141 | 0        | 1    | 4     | 0.0038 | 0        |
|   | 5000  | $x_3^4$ | 1    | 3     | 0.0036 | 2.22E-16 | 76   | 167   | 0.0841 | 9.12E-11     | 17   | 35    | 0.0213 | 7.89E-11 | 30   | 33    | 0.0331 | 5.60E-11 | 7    | 15    | 0.0223 | 8.18E-11 | 6    | 12    | 0.0102 | 5.11E-13 |
|   | 5000  | $x_4^4$ | 3    | 6     | 0.0055 | 0        | 82   | 170   | 0.0878 | 7.48E-11     | 23   | 50    | 0.0263 | 4.62E-11 | 38   | 44    | 0.0491 | 7.96E-11 | 17   | 27    | 0.0223 | 7.45E-12 | 5    | 9     | 0.0079 | 0        |
|   | 5000  | $x_5^4$ | 1    | 3     | 0.0035 | 0        | 78   | 159   | 0.0883 | 8.72E-11     | 20   | 43    | 0.0235 | 9.34E-11 | 37   | 42    | 0.0465 | 6.23E-11 | 8    | 12    | 0.0112 | 0        | 1    | 3     | 0.0033 | 0        |
|   | 5000  | $x_6^4$ | 3    | 6     | 0.0057 | 0        | 81   | 168   | 0.0840 | 9.95E-11     | 23   | 50    | 0.0262 | 5.00E-11 | 38   | 44    | 0.0458 | 8.14E-11 | 17   | 27    | 0.0219 | 7.35E-12 | 5    | 9     | 0.0090 | 0        |
|   | 10000 | $x_1^4$ | 1    | 3     | 0.0038 | 0        | 63   | 131   | 0.0978 | 9.34E-11     | 20   | 41    | 0.0344 | 9.77E-11 | 5    | 13    | 0.0144 | 0        | 5    | 13    | 0.0136 | 0        | 5    | 9     | 0.0118 | 5.96E-11 |
|   | 10000 | $x_2^4$ | 1    | 4     | 0.0051 | 0        | 83   | 173   | 0.1390 | 9.35E-11     | 2    | 6     | 0.0056 | 0        | 29   | 36    | 0.0520 | 0        | 10   | 16    | 0.0231 | 0        | 1    | 4     | 0.0057 | 0        |
|   | 10000 | $x_3^4$ | 1    | 3     | 0.0044 | 2.22E-16 | 77   | 169   | 0.1246 | 9.20E-11     | 17   | 35    | 0.0292 | 7.89E-11 | 30   | 33    | 0.0467 | 5.60E-11 | 7    | 15    | 0.0151 | 8.18E-11 | 6    | 12    | 0.0143 | 7.22E-13 |
|   | 10000 | $x_4^4$ | 3    | 6     | 0.0081 | 0        | 83   | 172   | 0.1284 | 7.45E-11     | 23   | 50    | 0.0381 | 6.63E-11 | 36   | 42    | 0.0637 | 5.33E-11 | 18   | 28    | 0.0327 | 5.63E-12 | 5    | 9     | 0.0109 | 0        |
|   | 10000 | $x_5^4$ | 1    | 3     | 0.0047 | 0        | 79   | 161   | 0.1275 | 8.80E-11     | 21   | 43    | 0.0345 | 9.25E-11 | 23   | 29    | 0.0420 | 0        | 27   | 42    | 0.0512 | 0        | 1    | 3     | 0.0045 | 0        |
|   | 10000 | $x_6^4$ | 3    | 6     | 0.0089 | 0        | 83   | 172   | 0.1360 | 7.25E-11     | 23   | 50    | 0.0396 | 6.90E-11 | 39   | 44    | 0.0638 | 9.73E-11 | 18   | 28    | 0.0347 | 5.67E-12 | 5    | 9     | 0.0110 | 0        |
|   | 50000 | $x_1^4$ | 1    | 3     | 0.0109 | 0        | 63   | 131   | 0.3748 | 9.40E-11     | 20   | 43    | 0.1409 | 8.74E-11 | 11   | 18    | 0.0832 | 0        | 5    | 13    | 0.0478 | 0        | 5    | 9     | 0.0395 | 5.96E-11 |
|   | 50000 | $x_2^4$ | 1    | 4     | 0.0134 | 0        | 86   | 179   | 0.5109 | 7.60E-11     | 2    | 6     | 0.0168 | 0        | 26   | 34    | 0.1714 | 0        | 13   | 23    | 0.1001 | 0        | 1    | 4     | 0.0141 | 0        |
|   | 50000 | $x_3^4$ | 1    | 3     | 0.0114 | 2.22E-16 | 80   | 175   | 0.5556 | 7.48E-11     | 17   | 35    | 0.1198 | 7.89E-11 | 30   | 33    | 0.1650 | 5.60E-11 | 7    | 15    | 0.0573 | 8.18E-11 | 6    | 12    | 0.0543 | 1.61E-12 |
|   | 50000 | $x_4^4$ | 3    | 6     | 0.0253 | 0        | 85   | 176   | 0.8151 | 8.39E-11     | 24   | 52    | 0.1478 | 3.43E-11 | 41   | 46    | 0.2328 | 5.89E-11 | 18   | 28    | 0.1272 | 2.04E-11 | 5    | 9     | 0.0410 | 0        |
|   | 50000 | $x_5^4$ | 1    | 3     | 0.0124 | 0        | 82   | 167   | 0.4907 | 7.15E-11     | 21   | 45    | 0.1333 | 8.27E-11 | 31   | 37    | 0.1950 | 9.83E-11 | 28   | 42    | 0.2025 | 0        | 1    | 3     | 0.0118 | 0        |
|   | 50000 | $x_6^4$ | 3    | 6     | 0.0264 | 0        | 85   | 176   | 0.5812 | 8.35E-11     | 24   | 52    | 0.1574 | 3.46E-11 | 41   | 46    | 0.2343 | 5.89E-11 | 18   | 28    | 0.1285 | 2.05E-11 | 5    | 9     | 0.0377 | 0        |
| 3 | 5000  | $x_1^4$ | 1    | 4     | 0.0050 | 0        | 75   | 376   | 0.1340 | 9.32E-11     | 12   | 40    | 0.0203 | 4.61E-11 | 1    | 6     | 0.0150 | 0        | 1    | 6     | 0.0049 | 0        | 5    | 11    | 0.0100 | 0        |
|   | 5000  | $x_2^4$ | 3    | 8     | 0.0072 | 0        | 89   | 448   | 0.1610 | 8.89E-11     | 14   | 44    | 0.0210 | 5.99E-11 | 74   | 140   | 0.1142 | 9.41E-11 | 14   | 49    | 0.0326 | 1.17E-11 | 1    | 4     | 0.0046 | 0        |
|   | 5000  | $x_3^4$ | 5    | 12    | 0.0093 | 4.44E-16 | 41   | 216   | 0.0797 | 7.65E-11     | 14   | 42    | 0.0217 | 1.78E-11 | 41   | 46    | 0.0523 | 4.13E-11 | 10   | 45    | 0.0260 | 1.62E-11 | 8    | 16    | 0.0147 | 0        |
|   | 5000  | $x_4^4$ | 1    | 4     | 0.0038 | 0        | 86   | 432   | 0.1558 | 8.03E-11     | 16   | 50    | 0.0238 | 5.32E-11 | 69   | 125   | 0.1098 | 6.40E-11 | 21   | 59    | 0.0413 | 2.03E-11 | 1    | 4     | 0.0038 | 0        |
|   | 5000  | $x_5^4$ | 1    | 4     | 0.0045 | 0        | 78   | 392   | 0.1439 | 8.80E-11     | 14   | 43    | 0.0218 | 5.10E-11 | 58   | 105   | 0.0918 | 6.46E-11 | 9    | 41    | 0.0247 | 8.67E-11 | 1    | 4     | 0.0041 | 0        |
|   | 5000  | $x_6^4$ | 1    | 4     | 0.0040 | 0        | 86   | 432   | 0.1568 | 8.06E-11     | 16   | 50    | 0.0236 | 5.33E-11 | 53   | 102   | 0.0883 | 7.35E-11 | 17   | 56    | 0.0392 | 1.28E-11 | 1    | 4     | 0.0038 | 0        |
|   | 10000 | $x_1^4$ | 1    | 4     | 0.0050 | 0        | 75   | 376   | 0.2040 | 9.32E-11     | 12   | 40    | 0.0300 | 6.35E-11 | 1    | 6     | 0.0063 | 0        | 1    | 6     | 0.0064 | 0        | 5    | 11    | 0.0145 | 0        |
|   | 10000 | $x_2^4$ | 3    | 8     | 0.0113 | 0        | 90   | 453   | 0.2483 | 8.91E-11     | 14   | 44    | 0.0339 | 8.45E-11 | 55   | 114   | 0.1485 | 9.88E-11 | 14   | 49    | 0.0545 | 8.20E-12 | 1    | 4     | 0.0054 | 0        |
|   | 10000 | $x_3^4$ | 5    | 12    | 0.0145 | 4.44E-16 | 5    | 53    | 0.0298 | 0            | 14   | 42    | 0.0341 | 1.78E-11 | 41   | 46    | 0.0785 | 4.13E-11 | 10   | 45    | 0.0401 | 1.62E-11 | 8    | 16    | 0.0221 | 0        |
|   | 10000 | $x_4^4$ | 1    | 4     | 0.0050 | 0        | 87   | 437   | 0.2446 | 8.43E-11     | 16   | 50    | 0.0393 | 7.53E-11 | 77   | 149   | 0.1850 | 9.7      |      |       |        |          |      |       |        |          |

Table 4: Results of test examples 4-6

| N     | Pdim  | DPRP |      |        | RESHYBRIDSCG |          |          | RDFPM1 |          |          | RDFPM2   |        |          | DFDPP    |          |        |          |          |          |        |          |          |          |        |          |          |
|-------|-------|------|------|--------|--------------|----------|----------|--------|----------|----------|----------|--------|----------|----------|----------|--------|----------|----------|----------|--------|----------|----------|----------|--------|----------|----------|
|       |       | IN   | FVAL | NORM   | IN           | FVAL     | PTIME    | NORM   | IN       | FVAL     | PTIME    | NORM   | IN       | FVAL     | PTIME    | NORM   |          |          |          |        |          |          |          |        |          |          |
| 4     | 5000  | x1   | 1    | 2      | 0.0044       | 0        | 67       | 132    | 0.00972  | 8.23E-11 | 2        | 3      | 0.0043   | 0        | 2        | 3      | 0.0048   | 0        | 2        | 4      | 0.0049   | 0        |          |        |          |          |
|       | 5000  | x2   | 2    | 3      | 0.0048       | 0        | 76       | 148    | 0.0089   | 8.39E-11 | 4        | 6      | 0.0066   | 0        | 29       | 31     | 0.0403   | 7.31E-11 | 9        | 10     | 0.0151   | 0        |          |        |          |          |
|       | 5000  | x3   | 3    | 0.0045 | 2.22E-16     | 5        | 6        | 0.0074 | 0        | 16       | 29       | 0.0214 | 9.29E-11 | 25       | 26       | 0.0344 | 1.19E-12 | 6        | 7        | 0.0101 | 8.87E-13 | 3        | 7        | 0.0080 | 0        |          |
|       | 5000  | x4   | 5    | 8      | 0.0095       | 7.78E-23 | 73       | 136    | 0.0036   | 9.25E-11 | 19       | 36     | 0.0346   | 7.77E-11 | 26       | 27     | 0.0366   | 0        | 10       | 11     | 0.0154   | 5.82E-13 | 2        | 4      | 0.0049   | 0        |
|       | 5000  | x5   | 8    | 3      | 0.0048       | 0        | 82       | 155    | 0.1124   | 8.72E-11 | 3        | 5      | 0.0059   | 0        | 21       | 22     | 0.0315   | 0        | 8        | 9      | 0.0133   | 1.53E-21 | 2        | 4      | 0.0057   | 0        |
|       | 5000  | x6   | 5    | 8      | 0.0098       | 7.79E-23 | 73       | 136    | 0.0050   | 8.12E-11 | 19       | 36     | 0.0323   | 7.78E-11 | 27       | 28     | 0.0373   | 0        | 10       | 11     | 0.0162   | 9.10E-13 | 2        | 4      | 0.0052   | 0        |
|       | 10000 | x1   | 1    | 2      | 0.0040       | 0        | 67       | 132    | 0.1587   | 8.32E-11 | 2        | 3      | 0.0064   | 0        | 3        | 4      | 0.0083   | 0        | 3        | 4      | 0.0098   | 0        | 2        | 4      | 0.0075   | 0        |
|       | 10000 | x2   | 3    | 0.0073 | 0            | 77       | 150      | 0.1668 | 8.59E-11 | 4        | 6        | 0.0123 | 0        | 21       | 22       | 0.0544 | 0        | 10       | 11       | 0.0279 | 4.32E-22 | 2        | 4        | 0.0084 | 0        |          |
|       | 10000 | x3   | 6    | 9      | 0.0166       | 3.14E-16 | 5        | 6      | 0.0121   | 0        | 16       | 31     | 0.0422   | 4.50E-11 | 25       | 26     | 0.0575   | 3.20E-12 | 7        | 8      | 0.0193   | 2.69E-12 | 3        | 7      | 0.0128   | 0        |
|       | 10000 | x4   | 5    | 8      | 0.0148       | 2.82E-23 | 71       | 137    | 0.1547   | 8.25E-11 | 19       | 38     | 0.0492   | 5.66E-11 | 25       | 26     | 0.0641   | 0        | 11       | 12     | 0.0311   | 8.99E-12 | 2        | 4      | 0.0077   | 0        |
|       | 10000 | x5   | 2    | 3      | 0.0076       | 0        | 83       | 157    | 0.1867   | 9.05E-11 | 3        | 5      | 0.0096   | 0        | 16       | 17     | 0.0379   | 0        | 8        | 9      | 0.0250   | 0        | 2        | 4      | 0.0076   | 0        |
|       | 10000 | x6   | 5    | 8      | 0.0160       | 2.82E-23 | 70       | 130    | 0.1562   | 9.50E-11 | 19       | 38     | 0.0470   | 5.67E-11 | 24       | 25     | 0.0621   | 0        | 11       | 12     | 0.0295   | 1.03E-11 | 2        | 4      | 0.0076   | 0        |
|       | 50000 | x1   | 1    | 2      | 0.0124       | 0        | 67       | 132    | 0.5398   | 8.39E-11 | 2        | 3      | 0.0190   | 0        | 3        | 4      | 0.0313   | 0        | 4        | 5      | 0.0428   | 0        | 2        | 4      | 0.0236   | 0        |
|       | 50000 | x2   | 3    | 0.0212 | 0            | 79       | 154      | 0.6541 | 9.89E-11 | 4        | 6        | 0.0360 | 0        | 26       | 27       | 0.2532 | 0        | 11       | 12       | 0.1102 | 5.39E-25 | 2        | 4        | 0.0267 | 0        |          |
|       | 50000 | x3   | 6    | 9      | 0.0567       | 3.14E-16 | 5        | 6      | 0.0384   | 0        | 16       | 31     | 0.1414   | 5.24E-11 | 24       | 25     | 0.1924   | 1.54E-11 | 7        | 8      | 0.0646   | 1.76E-11 | 3        | 7      | 0.0419   | 0        |
|       | 50000 | x4   | 5    | 8      | 0.0533       | 2.59E-24 | 78       | 146    | 0.6267   | 9.30E-11 | 20       | 40     | 0.1845   | 4.35E-11 | 24       | 25     | 0.2207   | 0        | 14       | 15     | 0.1341   | 9.02E-11 | 2        | 4      | 0.0249   | 0        |
|       | 50000 | x5   | 2    | 3      | 0.0218       | 0        | 85       | 161    | 0.6888   | 9.13E-11 | 3        | 5      | 0.0283   | 0        | 16       | 17     | 0.1483   | 0        | 10       | 11     | 0.1077   | 2.00E-25 | 2        | 4      | 0.0257   | 0        |
|       | 50000 | x6   | 5    | 8      | 0.0549       | 2.60E-24 | 78       | 146    | 0.6347   | 9.11E-11 | 20       | 40     | 0.1766   | 4.35E-11 | 25       | 26     | 0.2413   | 0        | 13       | 15     | 0.1319   | 4.40E-15 | 2        | 4      | 0.0243   | 0        |
|       | 5     | 5000 | x1   | 1      | 3            | 0.0048   | 0        | 39     | 80       | 0.0094   | 7.42E-11 | 21     | 43       | 0.0305   | 7.75E-11 | 11     | 17       | 0.0281   | 0        | 10     | 0.0103   | 7.93E-21 | 1        | 3      | 0.0033   | 0        |
|       |       | 5000 | x2   | 1      | 3            | 0.0039   | 0        | 41     | 85       | 0.0701   | 8.44E-11 | 21     | 45       | 0.0250   | 4.97E-11 | 20     | 29       | 0.0292   | 4.95E-25 | 12     | 20       | 0.0220   | 2.65E-21 | 1      | 3        | 0.0036   |
| 5000  |       | x3   | 1    | 3      | 0.0034       | 0        | 37       | 80     | 0.0478   | 5.88E-11 | 17       | 37     | 0.0246   | 5.42E-11 | 14       | 19     | 0.0203   | 7.40E-24 | 2        | 4      | 0.0051   | 0        | 6        | 12     | 0.0117   | 0        |
| 5000  |       | x4   | 1    | 3      | 0.0034       | 0        | 42       | 86     | 0.0529   | 7.32E-11 | 21       | 43     | 0.0280   | 7.85E-11 | 21       | 26     | 0.0304   | 2.36E-22 | 11       | 16     | 0.0185   | 0        | 1        | 3      | 0.0034   | 0        |
| 5000  |       | x5   | 1    | 3      | 0.0036       | 0        | 25       | 52     | 0.0363   | 4.28E-11 | 21       | 43     | 0.0295   | 7.70E-11 | 13       | 19     | 0.0203   | 0        | 8        | 16     | 0.0168   | 0        | 1        | 3      | 0.0043   | 0        |
| 5000  |       | x6   | 1    | 3      | 0.0035       | 0        | 32       | 65     | 0.0396   | 7.93E-11 | 21       | 43     | 0.0321   | 7.85E-11 | 23       | 29     | 0.0319   | 0        | 11       | 16     | 0.0183   | 1.95E-22 | 1        | 3      | 0.0040   | 0        |
| 10000 |       | x1   | 1    | 3      | 0.0041       | 0        | 42       | 85     | 0.0830   | 4.10E-11 | 21       | 45     | 0.0445   | 4.38E-11 | 6        | 12     | 0.0175   | 0        | 5        | 12     | 0.0183   | 1.68E-21 | 1        | 3      | 0.0043   | 0        |
| 10000 |       | x2   | 1    | 3      | 0.0045       | 0        | 43       | 89     | 0.0812   | 2.12E-11 | 21       | 45     | 0.0445   | 7.03E-11 | 22       | 31     | 0.0505   | 8.16E-25 | 10       | 13     | 0.0232   | 0        | 1        | 3      | 0.0049   | 0        |
| 10000 |       | x3   | 1    | 3      | 0.0048       | 0        | 37       | 80     | 0.0740   | 8.32E-11 | 17       | 37     | 0.0357   | 5.42E-11 | 14       | 19     | 0.0324   | 7.46E-24 | 2        | 4      | 0.0071   | 0        | 6        | 12     | 0.0156   | 0        |
| 10000 |       | x4   | 1    | 3      | 0.0044       | 0        | 45       | 90     | 0.0867   | 6.19E-11 | 21       | 45     | 0.0443   | 4.44E-11 | 19       | 24     | 0.0419   | 6.81E-22 | 10       | 12     | 0.0223   | 0        | 1        | 3      | 0.0051   | 0        |
| 10000 |       | x5   | 1    | 3      | 0.0046       | 0        | 25       | 52     | 0.0486   | 6.05E-11 | 21       | 45     | 0.0497   | 4.36E-11 | 14       | 19     | 0.0330   | 9.96E-20 | 7        | 14     | 0.0208   | 2.43E-20 | 1        | 3      | 0.0047   | 0        |
| 10000 |       | x6   | 1    | 3      | 0.0042       | 0        | 44       | 90     | 0.0902   | 9.70E-12 | 21       | 45     | 0.0440   | 4.44E-11 | 22       | 28     | 0.0546   | 0        | 10       | 12     | 0.0218   | 0        | 1        | 3      | 0.0047   | 0        |
| 50000 |       | x1   | 1    | 3      | 0.0127       | 0        | 44       | 91     | 0.3348   | 4.59E-11 | 21       | 45     | 0.1795   | 9.80E-11 | 7        | 12     | 0.0737   | 3.70E-21 | 3        | 8      | 0.0411   | 3.57E-18 | 1        | 3      | 0.0130   | 0        |
| 50000 |       | x2   | 1    | 3      | 0.0145       | 0        | 43       | 89     | 0.3117   | 4.79E-11 | 22       | 47     | 0.1912   | 4.40E-11 | 14       | 21     | 0.1363   | 7.60E-19 | 14       | 22     | 0.1409   | 8.91E-20 | 1        | 3      | 0.0164   | 0        |
| 50000 |       | x3   | 1    | 3      | 0.0140       | 0        | 39       | 84     | 0.2941   | 6.34E-11 | 17       | 37     | 0.1446   | 5.42E-11 | 14       | 21     | 0.1333   | 7.40E-24 | 2        | 4      | 0.0221   | 0        | 6        | 12     | 0.0071   | 0        |
| 50000 |       | x4   | 1    | 3      | 0.0159       | 0        | 43       | 88     | 0.3220   | 8.24E-11 | 21       | 45     | 0.1744   | 9.93E-11 | 29       | 34     | 0.2466   | 0        | 12       | 14     | 0.1044   | 0        | 1        | 3      | 0.0135   | 0        |
| 50000 |       | x5   | 1    | 3      | 0.0138       | 0        | 27       | 56     | 0.2967   | 6.27E-11 | 21       | 45     | 0.1822   | 9.74E-11 | 24       | 30     | 0.2027   | 9.48E-11 | 9        | 15     | 0.0943   | 7.19E-18 | 1        | 3      | 0.0142   | 0        |
| 50000 |       | x6   | 1    | 3      | 0.0137       | 0        | 41       | 84     | 0.2981   | 1.84E-11 | 21       | 45     | 0.2024   | 9.93E-11 | 19       | 26     | 0.1773   | 2.30E-23 | 12       | 14     | 0.1066   | 0        | 1        | 3      | 0.0138   | 0        |
| 6     |       | 5000 | x1   | 6      | 13           | 0.0260   | 1.85E-11 | 9      | 19       | 0.0161   | 5.03E-11 | 22     | 47       | 0.0428   | 4.34E-11 | 11     | ***      | ***      | ***      | ***    | ***      | ***      | 11       | 23     | 0.0289   | 1.61E-11 |
|       |       | 5000 | x2   | 6      | 13           | 0.0159   | 1.49E-11 | 10     | 21       | 0.0091   | 7.48E-11 | 22     | 45       | 0.0469   | 8.75E-11 | 11     | ***      | ***      | ***      | ***    | ***      | ***      | 11       | 23     | 0.0268   | 1.00E-11 |
|       | 5000  | x3   | 6    | 13     | 0.0159       | 2.27E-11 | 15       | 31     | 0.0266   | 8.63E-11 | 22       | 47     | 0.0440   | 5.32E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 10       | 21       | 0.0249 | 8.74E-11 |          |
|       | 5000  | x4   | 6    | 13     | 0.0155       | 1.89E-11 | 9        | 19     | 0.0169   | 9.84E-11 | 22       | 47     | 0.0441   | 4.38E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0298 | 1.33E-11 |          |
|       | 5000  | x5   | 6    | 13     | 0.0155       | 1.89E-11 | 9        | 19     | 0.0162   | 5.49E-11 | 22       | 47     | 0.0436   | 4.37E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0290 | 1.33E-11 |          |
|       | 5000  | x6   | 6    | 13     | 0.0161       | 1.86E-11 | 9        | 19     | 0.0151   | 9.84E-11 | 22       | 47     | 0.0462   | 4.38E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0271 | 1.33E-11 |          |
|       | 10000 | x1   | 6    | 13     | 0.0246       | 2.61E-11 | 9        | 19     | 0.0274   | 7.12E-11 | 22       | 47     | 0.0713   | 6.14E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0451 | 2.28E-11 |          |
|       | 10000 | x2   | 6    | 13     | 0.0247       | 2.10E-11 | 9        | 19     | 0.0282   | 6.38E-11 | 22       | 47     | 0.0726   | 4.95E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0509 | 1.50E-11 |          |
|       | 10000 | x3   | 6    | 13     | 0.0260       | 3.20E-11 | 10       | 21     | 0.0339   | 4.73E-11 | 22       | 47     | 0.0737   | 7.52E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0459 | 1.03E-11 |          |
|       | 10000 | x4   | 6    | 13     | 0.0262       | 2.63E-11 | 9        | 19     | 0.0281   | 6.50E-11 | 22       | 47     | 0.0786   | 6.19E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0486 | 1.88E-11 |          |
|       | 10000 | x5   | 6    | 13     | 0.0254       | 2.63E-11 | 9        | 19     | 0.0291   | 5.98E-11 | 22       | 47     | 0.0805   | 6.17E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0462 | 1.87E-11 |          |
|       | 10000 | x6   | 6    | 13     | 0.0251       | 2.63E-11 | 9        | 19     | 0.0270   | 6.50E-11 | 22       | 47     | 0.0701   | 6.19E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0465 | 1.88E-11 |          |
|       | 50000 | x1   | 6    | 13     | 0.1121       | 5.89E-11 | 10       | 21     | 0.1433   | 6.33E-12 | 23       | 47     | 0.3262   | 9.61E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0465 | 1.88E-11 |          |
|       | 50000 | x2   | 6    | 13     | 0.1079       | 4.71E-11 | 10       | 21     | 0.1305   | 4.27E-12 | 23       | 47     | 0.3196   | 7.75E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0465 | 1.88E-11 |          |
|       | 50000 | x3   | 6    | 13     | 0.1170       | 7.15E-11 | 9        | 19     | 0.1210   | 4.71E-11 | 23       | 49     | 0.3241   | 4.71E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0465 | 1.88E-11 |          |
|       | 50000 | x4   | 6    | 13     | 0.1085       | 5.88E-11 | 10       | 21     | 0.1373   | 5.27E-12 | 23       | 47     | 0.2973   | 9.69E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0462 | 1.87E-11 |          |
|       | 50000 | x5   | 6    | 13     | 0.1096       | 5.87E-11 | 11       | 22     | 0.1415   | 8.18E-11 | 23       | 47     | 0.3075   | 9.66E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0465 | 1.88E-11 |          |
|       | 50000 | x6   | 6    | 13     | 0.1075       | 5.85E-11 | 10       | 21     | 0.1315   | 5.27E-12 | 23       | 47     | 0.3184   | 9.69E-11 | 11       | ***    | ***      | ***      | ***      | ***    | ***      | 11       | 23       | 0.0462 | 1.87E-11 |          |

Table 5: Results of test examples 7-8

| N     | Pdim    | DFPRP   |      |        |          | MPRP1    |      |        |          | RESHYBRIDSCG |      |        |          | RDFPM1   |      |        |          | RDFPM2   |      |        |          | DFDFP    |      |        |          |          |
|-------|---------|---------|------|--------|----------|----------|------|--------|----------|--------------|------|--------|----------|----------|------|--------|----------|----------|------|--------|----------|----------|------|--------|----------|----------|
|       |         | IN      | FVAL | PTIME  | NORM     | IN       | FVAL | PTIME  | NORM     | IN           | FVAL | PTIME  | NORM     | IN       | FVAL | PTIME  | NORM     | IN       | FVAL | PTIME  | NORM     | IN       | FVAL | PTIME  | NORM     |          |
| 7     | 5000    | $x_1^8$ | 8    | 15     | 0.0196   | 2.22E-12 | 65   | 308    | 0.1337   | 9.30E-11     | 14   | 46     | 0.0306   | 4.12E-11 | 11   | 34     | 0.0397   | 6.75E-11 | 11   | 49     | 0.0360   | 5.38E-11 | 13   | 27     | 0.0265   | 1.11E-11 |
|       | 5000    | $x_2^8$ | 10   | 18     | 0.0188   | 6.04E-11 | 62   | 307    | 0.1423   | 9.11E-11     | 17   | 51     | 0.0358   | 7.03E-11 | 25   | 56     | 0.0570   | 7.32E-11 | 15   | 60     | 0.0515   | 3.01E-11 | 12   | 29     | 0.0253   | 2.60E-11 |
|       | 5000    | $x_3^8$ | 13   | 24     | 0.0247   | 6.16E-12 | 70   | 341    | 0.1394   | 9.87E-11     | 16   | 52     | 0.0341   | 9.93E-11 | 37   | 70     | 0.0835   | 8.46E-11 | 14   | 58     | 0.0436   | 5.12E-11 | 13   | 27     | 0.0266   | 5.01E-11 |
|       | 5000    | $x_4^8$ | 17   | 30     | 0.0346   | 1.91E-12 | 66   | 330    | 0.1398   | 9.85E-11     | 16   | 52     | 0.0343   | 4.32E-11 | 31   | 74     | 0.0818   | 4.65E-11 | 16   | 63     | 0.0508   | 3.87E-11 | 13   | 27     | 0.0262   | 5.23E-11 |
|       | 5000    | $x_5^8$ | 13   | 22     | 0.0238   | 8.99E-11 | 65   | 325    | 0.1328   | 9.90E-11     | 16   | 52     | 0.0340   | 3.54E-11 | 34   | 67     | 0.0818   | 7.37E-11 | 17   | 63     | 0.0560   | 3.24E-11 | 12   | 25     | 0.0243   | 6.36E-11 |
|       | 5000    | $x_6^8$ | 17   | 30     | 0.0312   | 1.75E-12 | 66   | 330    | 0.1394   | 9.86E-11     | 16   | 52     | 0.0342   | 4.31E-11 | 31   | 78     | 0.0808   | 6.42E-11 | 16   | 63     | 0.0506   | 3.89E-11 | 13   | 27     | 0.0262   | 5.22E-11 |
|       | 10000   | $x_1^8$ | 8    | 15     | 0.0255   | 3.14E-12 | 62   | 300    | 0.2088   | 9.92E-11     | 14   | 46     | 0.0488   | 5.83E-11 | 11   | 34     | 0.0498   | 9.51E-11 | 11   | 49     | 0.0583   | 7.60E-11 | 13   | 27     | 0.0412   | 1.58E-11 |
|       | 10000   | $x_2^8$ | 10   | 18     | 0.0290   | 8.54E-11 | 63   | 312    | 0.1980   | 8.57E-11     | 17   | 51     | 0.0573   | 9.95E-11 | 19   | 52     | 0.0805   | 3.56E-11 | 15   | 57     | 0.0727   | 8.01E-11 | 12   | 29     | 0.0398   | 3.20E-11 |
|       | 10000   | $x_3^8$ | 17   | 30     | 0.0510   | 7.30E-12 | 71   | 346    | 0.2229   | 8.15E-11     | 17   | 52     | 0.0559   | 8.24E-11 | 30   | 64     | 0.1113   | 7.46E-11 | 15   | 61     | 0.0773   | 4.86E-11 | 13   | 27     | 0.0398   | 7.09E-11 |
|       | 10000   | $x_4^8$ | 17   | 30     | 0.0530   | 2.64E-12 | 67   | 335    | 0.2398   | 9.26E-11     | 16   | 52     | 0.0618   | 6.10E-11 | 34   | 75     | 0.1712   | 6.39E-11 | 18   | 65     | 0.0895   | 6.45E-11 | 13   | 27     | 0.0416   | 7.39E-11 |
|       | 10000   | $x_5^8$ | 13   | 24     | 0.0397   | 9.78E-13 | 66   | 330    | 0.2074   | 9.31E-11     | 16   | 52     | 0.0548   | 5.01E-11 | 31   | 68     | 0.1343   | 9.50E-11 | 19   | 65     | 0.0850   | 3.32E-11 | 12   | 25     | 0.0439   | 9.00E-11 |
|       | 10000   | $x_6^8$ | 17   | 30     | 0.0527   | 2.53E-12 | 67   | 335    | 0.2265   | 9.27E-11     | 16   | 52     | 0.0542   | 6.10E-11 | 38   | 84     | 0.1606   | 4.77E-11 | 18   | 65     | 0.0889   | 6.60E-11 | 13   | 27     | 0.0395   | 7.39E-11 |
|       | 50000   | $x_1^8$ | 8    | 15     | 0.1056   | 7.01E-12 | 62   | 306    | 0.7964   | 7.34E-11     | 15   | 46     | 0.2002   | 7.64E-11 | 12   | 31     | 0.1944   | 5.80E-11 | 12   | 53     | 0.2862   | 2.55E-11 | 13   | 27     | 0.1672   | 3.53E-11 |
|       | 50000   | $x_2^8$ | 10   | 20     | 0.1258   | 1.47E-12 | 65   | 322    | 0.8268   | 8.47E-11     | 17   | 54     | 0.2402   | 7.25E-11 | 21   | 53     | 0.3287   | 8.71E-11 | 16   | 61     | 0.3144   | 2.69E-11 | 12   | 29     | 0.1789   | 7.14E-11 |
|       | 50000   | $x_3^8$ | 18   | 30     | 0.1884   | 5.02E-11 | 73   | 356    | 0.9277   | 8.06E-11     | 17   | 55     | 0.2349   | 6.01E-11 | 28   | 68     | 0.4228   | 9.44E-11 | 15   | 58     | 0.3008   | 2.56E-11 | 14   | 29     | 0.1816   | 1.40E-11 |
|       | 50000   | $x_4^8$ | 17   | 30     | 0.2065   | 5.81E-12 | 69   | 345    | 0.9312   | 9.16E-11     | 17   | 52     | 0.2196   | 8.00E-11 | 30   | 70     | 0.4638   | 5.87E-11 | 19   | 69     | 0.3723   | 2.19E-11 | 14   | 29     | 0.1846   | 1.47E-11 |
| 50000 | $x_5^8$ | 13      | 24   | 0.1522 | 2.19E-12 | 68       | 340  | 0.8824 | 9.21E-11 | 17           | 52   | 0.2421 | 6.57E-11 | 28       | 63   | 0.4261 | 9.65E-11 | 22       | 69   | 0.3825 | 1.79E-11 | 13       | 27   | 0.1739 | 1.78E-11 |          |
| 50000 | $x_6^8$ | 17      | 30   | 0.2020 | 5.76E-12 | 69       | 345  | 0.9193 | 9.16E-11 | 17           | 52   | 0.2305 | 8.00E-11 | 44       | 90   | 0.6693 | 7.39E-11 | 19       | 69   | 0.3892 | 2.20E-11 | 14       | 29   | 0.1961 | 1.47E-11 |          |
| 8     | 5000    | $x_1^8$ | 11   | 23     | 0.0356   | 8.65E-11 | 31   | 81     | 0.0591   | 4.89E-11     | 20   | 44     | 0.0389   | 4.99E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 14   | 26     | 0.0355   | 2.41E-11 |
|       | 5000    | $x_2^8$ | 11   | 25     | 0.0293   | 8.60E-11 | 30   | 77     | 0.0576   | 1.00E-10     | 19   | 41     | 0.0348   | 7.17E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 14   | 27     | 0.0316   | 1.49E-11 |
|       | 5000    | $x_3^8$ | 12   | 25     | 0.0320   | 4.63E-11 | 32   | 86     | 0.0614   | 9.90E-11     | 19   | 42     | 0.0435   | 2.98E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 13   | 25     | 0.0350   | 9.74E-11 |
|       | 5000    | $x_4^8$ | 11   | 25     | 0.0297   | 7.91E-11 | 31   | 83     | 0.0575   | 8.10E-11     | 20   | 42     | 0.0368   | 9.35E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 14   | 27     | 0.0315   | 2.89E-11 |
|       | 5000    | $x_5^8$ | 11   | 23     | 0.0281   | 7.22E-11 | 35   | 88     | 0.0648   | 7.52E-11     | 18   | 40     | 0.0348   | 9.46E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 13   | 25     | 0.0306   | 5.90E-11 |
|       | 5000    | $x_6^8$ | 11   | 25     | 0.0291   | 4.86E-11 | 37   | 97     | 0.0719   | 9.24E-11     | 19   | 42     | 0.0386   | 4.11E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 13   | 25     | 0.0299   | 6.64E-11 |
|       | 10000   | $x_1^8$ | 12   | 27     | 0.0500   | 6.04E-11 | 36   | 92     | 0.1210   | 1.00E-11     | 21   | 46     | 0.0666   | 4.70E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 15   | 28     | 0.0595   | 1.75E-11 |
|       | 10000   | $x_2^8$ | 11   | 23     | 0.0441   | 6.57E-11 | 31   | 77     | 0.0922   | 5.49E-11     | 18   | 40     | 0.0570   | 8.89E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 14   | 27     | 0.0535   | 1.54E-11 |
|       | 10000   | $x_3^8$ | 12   | 27     | 0.0537   | 3.16E-11 | 37   | 87     | 0.1095   | 4.92E-11     | 22   | 47     | 0.0751   | 4.70E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 11   | 22     | 0.0425   | 8.63E-11 |
|       | 10000   | $x_4^8$ | 12   | 27     | 0.0524   | 8.13E-11 | 38   | 93     | 0.1212   | 8.92E-12     | 20   | 44     | 0.0634   | 3.06E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 13   | 26     | 0.0502   | 1.82E-11 |
|       | 10000   | $x_5^8$ | 12   | 27     | 0.0634   | 6.58E-11 | 34   | 95     | 0.1076   | 8.57E-11     | 18   | 40     | 0.0577   | 5.61E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 17   | 30     | 0.0746   | 3.17E-11 |
|       | 10000   | $x_6^8$ | 12   | 27     | 0.0473   | 7.45E-11 | 46   | 114    | 0.1470   | 1.44E-11     | 21   | 46     | 0.0708   | 2.57E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 14   | 27     | 0.0544   | 1.06E-12 |
|       | 50000   | $x_1^8$ | 14   | 29     | 0.2371   | 3.13E-11 | 44   | 104    | 0.5748   | 6.22E-11     | 20   | 44     | 0.2871   | 9.14E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 16   | 30     | 0.2932   | 9.82E-12 |
|       | 50000   | $x_2^8$ | 13   | 29     | 0.2279   | 9.21E-11 | 57   | 129    | 0.7292   | 2.98E-11     | 21   | 45     | 0.2965   | 9.16E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 15   | 28     | 0.2866   | 2.49E-11 |
|       | 50000   | $x_3^8$ | 14   | 29     | 0.2464   | 3.77E-11 | 35   | 92     | 0.4736   | 8.80E-11     | 20   | 42     | 0.2591   | 7.21E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 16   | 30     | 0.2874   | 2.33E-11 |
|       | 50000   | $x_4^8$ | 14   | 29     | 0.2315   | 4.56E-11 | 42   | 110    | 0.5665   | 8.14E-11     | 21   | 46     | 0.2911   | 5.49E-11 | ***  | ***    | ***      | ***      | ***  | ***    | ***      | ***      | 13   | 25     | 0.2288   | 6.79E-11 |
| 50000 | $x_5^8$ | 14      | 29   | 0.2349 | 3.69E-11 | 55       | 145  | 0.7529 | 3.26E-11 | 20           | 42   | 0.2650 | 8.58E-11 | ***      | ***  | ***    | ***      | ***      | ***  | ***    | ***      | 16       | 29   | 0.2684 | 3.23E-11 |          |
| 50000 | $x_6^8$ | 14      | 29   | 0.2395 | 4.24E-11 | 34       | 89   | 0.4885 | 2.60E-11 | 21           | 46   | 0.2883 | 5.72E-11 | ***      | ***  | ***    | ***      | ***      | ***  | ***    | ***      | 16       | 29   | 0.2670 | 2.26E-11 |          |